

Home Bias and International Risk Sharing

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Abstract

This paper investigates the relationship between equity home bias and international risk sharing. The analysis is based on panel-data regressions using OECD national accounts data and foreign asset holdings.

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1 Introduction

Sørensen et al. argue in their 2007 paper *Home Bias and International Risk Sharing: Twin Puzzles Separated at Birth* that the macroeconomic concept of international risk sharing and the financial economic finding that investors tend to hold a disproportionately large share of domestic assets – the so-called home bias – are closely related and may be manifestations of the same underlying behavior.

We use Sørensen et al. (2007) as a starting point to investigate this relationship further, replicate their main findings, and extend their analysis by a time period of 20 years, as well as by a larger country sample of XX OECD countries.¹

We then depart from the original paper by shifting our focus to the euro area as an interesting case study for investigating the relationship between home bias and international risk sharing. European integration, both during the original paper’s sample period (1993–2003) and during our extended time period (1987–2017), provides an opportunity to analyze whether a decline in exchange rate risk and transaction costs is directly associated with a declining home bias and an increase in international risk sharing. Puzzling results from a naive analysis, which suggest that euro area countries became more diversified with countries *outside* the euro area during the process of integration, lead us to an extensive discussion of financial centers in the euro area. We identify a potential shortcoming of the original paper that may have become increasingly relevant over the past two decades due to the rise of financial centers within the euro area. The contribution of this extension should therefore be understood as a friendly warning to future research extending Sørensen et al. (2007), suggesting that this shortcoming may distort an uncorrected analysis.

Second, we use the euro area case study to focus on the channels of risk sharing. Combining Sørensen et al. (2007) with Asdrubali, Sorensen, and Yosha (1996), we note that risk sharing *within* the euro area has increased after 1999 and analyze whether this increase can be attributed specifically to a reduction in home bias. Using a variance decomposition method, we find that home bias and risk sharing are negatively associated in the euro area, confirming the findings of Sørensen et al. (2007) at a more granular level and with a different methodology.

2 Panel regressions

2.1 Theoretical outline

Sørensen et al. (2007) use a panel regression framework to analyse whether a lower equity home bias is associated with a higher degree of income risk sharing and consumption risk

¹Country sample extensions were constrained by data availability.

sharing. To do so, they regress the log growth rate of country-specific Gross National Income (GNI) or consumption (C), relative to the sample-wide log growth rate of GNI or consumption, on the country-specific log growth rate of Gross Domestic Product (GDP), relative to the sample-wide log growth rate of GDP. The baseline regression is

$$\Delta \log C_{it} - \Delta \log C_t = \alpha_i + \kappa (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it}. \quad (1)$$

As in the cross-sectional regressions, one minus the respective coefficients κ measures the average degree of risk sharing. A lower coefficient therefore corresponds to a higher degree of risk sharing.

To incorporate equity home bias into the panel regression, the authors allow the coefficient on relative GDP growth to vary with both a time trend and the deviation of a country's equity home bias from the sample average:

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(EHB_{it} - \overline{EHB}_t) \quad (2)$$

It is therefore most intuitive to interpret the implied degree of risk sharing directly. and for consumption risk sharing by

$$CRS_{it} = 1 - \kappa_0 - \kappa_1(t - \bar{t}) - \kappa_2(EHB_{it} - \overline{EHB}_t). \quad (3)$$

where CRS_{it} stand for income and consumption risk sharing. Similarly, they run the above regression with GNI plugged in for consumption.² For the hypothesis of a potential co-movement between equity home bias and risk sharing, the main coefficient of interest is therefore $-\kappa_2$. A positive value of κ_2 implies that a higher equity home bias increases the sensitivity of relative consumption growth to relative GDP growth and therefore reduces consumption risk sharing. Equivalently, an increase in equity home bias by 0.1 changes consumption risk sharing by $-0.1\kappa_2$. All panel regressions are estimated using a two-stage feasible GLS-type approach. In the second-stage regression, each country is assigned a weight proportional to one over the the standard error of its first-stage residuals.³

2.2 Empirical results

When reporting the results of the panel regressions, we follow the authors standard and report the easier to interpret coefficients $1 - \kappa_0$, $-\kappa_1$ and $-\kappa_2$ multiplied by 100 to interpret them as percentage effects. Our panel regression results follow the key results of Sørensen et al. (2007), with a statistically non-different to zero estimate for the time

²See Sørensen et al. (2007) for a more extensive discussion. For the sake of simplicity, we decided to present only the consumption regression. In the tables below, both regression results are reported.

³Since this procedure is weakly documented in the original paper, we compared the first- and second-stage regression results. The differences do not affect the direction or statistical significance of the main coefficients.

trend and a highly significant coefficient for the equity home bias both on income and consumption risk sharing. An increase in the equity home bias by 0.1 is now associated with an increase in income risk sharing by roughly 13 percent and consumption risk sharing by roughly 20 percent. Given that our point estimates for the coefficient are larger than 100 and notably higher than the point estimates found by Sørensen et al. (2007), we believe that the discussed data choices, as well as data revisions and the choice of the base year play an important role. Further, we rely on an interpretation offered by the authors themselves, believing that a coefficient of more than 100 percent risk sharing likely indicates a strong co-movement, even if the point estimate is not likely to hold in larger samples.⁴ Our coefficient for average income risk sharing is highly significant, while it is not in the original paper. Given that it will turn non-significant in the larger sample period, we observe this, but doubt that it affects the main findings of this exercise.

Table 1: Income and Consumption Risk Sharing and Equity Home Bias: OECD 1993–2003

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	16 (4.16)	1 (0.76)	−133 (4.87)
Consumption	55 (15.10)	1 (0.52)	−204 (9.07)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. The income and consumption row present $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

Running the same regressions on foreign asset holdings over GDP, as done by Sørensen et al. (2007) confirms the robustness of our replication exercise. Even though we find partly differing point estimates, the direction and significance of the point estimates are in line with the original paper.⁵ We treat these regressions as an robustness check and find positive $-\kappa_2$ coefficients for all forms of asset holdings, and statistically non significant estimates for the time trend.

Leading us to conclude that the main findings of Sørensen et al. (2007) panel regressions can be replicated with updated data two decades later. We treat these regressions merely as an robustness check, and find positive

⁴A statement that proves to be true once we extend the sample period from 1993–2003 to 1987–2017.

⁵Note that a positive sign of $-\kappa_2$ is now associated with a higher level of risk sharing.

Table 2: Risk Sharing and Foreign Asset Holdings Relative to GDP: OECD 1993–2003

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	15 (4.15)	1 (1.21)	12 (5.19)				
	16 (4.21)	1 (0.72)		18 (5.15)			
	10 (2.79)	2 (1.63)			12 (3.33)		
	16 (4.25)	1 (0.73)				18 (5.05)	
	16 (4.26)	1 (1.06)					21 (5.14)
Consumption risk sharing	57 (14.77)	1 (0.90)	18 (8.80)				
	50 (12.16)	0 (0.15)		21 (6.01)			
	52 (12.59)	0 (0.41)			14 (6.16)		
	52 (12.93)	0 (0.32)				23 (6.81)	
	55 (13.59)	0 (0.46)					24 (7.35)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

3 Extentions

3.1 First Extention: Similar Country Sample 1987–2017

By keeping the original sample of 22 OECD countries and extending the observation period by 20 years to 1987–2017 we aim to analyse whether the authors findings are robust on a longer time horizon, as the original sample period of 11 years is rather short.

FELIX TEIL: CROSS SECTION: Looking at figure (XXX), we can see that the cross section risk sharing coefficient has increased constantly from 1987 to 2000, using the same smoothing method as the authors in the paper. After a short dip during 2000 to 2003, potentially caused by the *dotcom*-bubble, we observe an even stronger increase in cross section risk sharing that is ended abruptly by the global financial crisis 2007/2008. After this crisis, global cross section risk sharing grows even stronger. Looking at the curve for foreign assets over GDP we can see that they constantly increased over the time period of 31 years.

Panel regressions find that the main relationship between Equity Home Bias and risk sharing largely hold when extending the time period by 20 years. A 0.1 decrease of the equity home bias is now associated with a roughly 3 percent increase in income risk sharing and a 8 percent increase in consumption risk sharing. The average risk sharing coefficient decreases notably, and becomes statistically non-significant for income risk sharing. Given that this coefficient was non significant in Sørensen et al. (2007) for the original sample period, it seems to be a parameter that varies, depending on sample and

data. Most relevant is the significance of $-\kappa_2$. / Focusing on the foreign asset over GDP regressions gives robustness to conclude that the negative relationship between Home Bias and risk sharing holds over the extended time period. conclude that the negative relationship between Home Bias and Income and Consumption Risk sharing holds for the extended period.

Table 3: Income and Consumption Risk Sharing and Equity Home Bias: OECD 1987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.14)	0 (0.82)	−28 (2.80)
Consumption	40 (19.61)	0 (0.29)	−81 (7.96)

Note: Values in parentheses are absolute t-values. Replication sample with 22 OECD countries included in the panel regression in table 3 of the original paper. Extended time period 1987–2017 For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

Table 4: Risk Sharing and Foreign Asset Holdings Relative to GDP: OECD 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	−2 (0.87)	0 (0.70)	5 (3.59)				
	0 (0.21)	0 (0.80)		8 (4.14)			
	−2 (0.98)	0 (0.40)			4 (2.57)		
	−1 (0.62)	0 (0.91)				8 (4.15)	
	−2 (1.03)	0 (0.74)					8 (4.08)
Consumption risk sharing	42 (20.31)	0 (0.28)	10 (9.07)				
	39 (18.10)	0 (0.72)		12 (5.60)			
	40 (19.21)	0 (0.18)			11 (7.41)		
	40 (18.98)	0 (0.58)				14 (7.06)	
	41 (19.35)	0 (0.40)					15 (7.60)

Note: Values in parentheses are absolute t-values. Original sample and extended time period. For income risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. For consumption risk sharing, the reported coefficients are $100 \cdot (1 - \eta_0)$, $100 \cdot (-\eta_1)$, and $100 \cdot (-\eta_2)$. Country-fixed effects included.

3.2 Second Extention: Extended Country Sample 1987–2017

Extending the OECD sampe to 29 countries ⁶ for the sample period 1987–2017 gives us the opportunity to analyse if the main findings of Sørensen et al. (2007) can be extended to a broader set of countries. As a potential shortcoming of this extention, we must admit that the sample size is still rather small. The panel-regression results of the (slightly) extended sample between 1987 and 2017 confirm the same results as shown in the two regressions before. An decrease in the Equity Home Bias by 0.1 is associated with an increase of income risk sharing by roughly 3 percent and an increase in consumption risk sharing by roughly 8 percent.

again, the panel regressions for foreign Asset holdings relative to GDP confirm this estimation, with highly significant estimations for the asset holdings themselves, and non significant estimations for the time trend. Again, the average risk sharing coefficient for income is non significant in both specifications, while it is highly significant for consumption in both specifications.

Table 5: Income and Consumption Risk Sharing and Equity Home Bias: Extended OECD Sample 19987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.34)	0 (0.26)	−28 (3.24)
Consumption	30 (15.28)	0 (0.41)	−84 (8.25)

Note: Values in parentheses are absolute t-values. Extended sample with up to 29 OECD countries, many observations lost due to missing data coverage. Extended time period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

4 Case Study Euro Area

Using the EA as its own sample leads to rather unintuitive results, both on a cross section and panel regression level. As can be seen in the following two tables, the point estimates are notably smaller than for the OECD sample and the coefficient for the Equity Home Bias on consumption is not significant anymore. Sørensen et al. (2007) note the same puzzling irregularities when looking at an EU sample, especially with respect to the Consumption risk sharing coefficient among EU countries. They cite (XXX) and

⁶For the GNI analysis and panel regressions some observations drop, due to a lack of data coverage for both Consumption, GNI, GDP and the Equity Home Bias. A future extention of Sørensen et al. (2007) should certainly try extend the country sample beyond OECD countries. Due to our data problems, we decided to keep a rather small, but consistent sample.

Table 6: Risk Sharing and Foreign Asset Holdings Relative to GDP: Extended OECD Sample and 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP				
		Trend	Equity	Debt	FDI	Equity + Debt All Assets
Income risk sharing	1 (0.63)	0 (0.68)	3 (3.40)			
	0 (0.20)	0 (0.93)		7 (4.72)		
	−2 (1.23)	0 (0.62)			4 (3.23)	
	0 (0.05)	0 (1.09)				7 (4.65)
	0 (0.27)	0 (0.94)				7 (4.54)
Consumption risk sharing	28 (14.42)	0 (0.56)	12 (11.26)			
	30 (14.02)	0 (0.49)		10 (4.75)		
	26 (11.85)	0 (2.00)			12 (7.78)	
	28 (13.53)	0 (0.00)				13 (6.73)
	28 (13.46)	0 (0.27)				14 (7.37)

Note: Values in parentheses are absolute t-values. Extended sample with up to 29 OECD countries, many observations lost due to missing data coverage. Extended time period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

speculate that consumption risk sharing within the EU has decreased due to less-pro cyclical government spending among EU countries due to the Maastricht criteria. We understand that this reasoning can be applied to the EA aswell and tried to run structural break regressions to understand this relationship further. Due non robustly significant results in those structural break regressions, we shift our focus to a different set of tools to analyse the relationship further.

Table 7: Income and Consumption Risk Sharing and Equity Home Bias: Euroarea 1987–2017

	Average Risk Sharing	Time Trend	Equity Home Bias
Income	0 (0.09)	0 (0.96)	−30 (2.16)
Consumption	27 (8.85)	−1 (3.48)	−8 (0.50)

Note: Values in parentheses are absolute t-values. Euroarea sample for the period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.

References

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Table 8: Risk Sharing and Foreign Asset Holdings Relative to GDP: Euroarea 1987–2017

With country-fixed effects	Average risk sharing	Interaction terms with GDP					
		Trend	Equity	Debt	FDI	Equity + Debt	All Assets
Income risk sharing	1 (0.15)	0 (0.17)	3 (0.86)				
	−7 (1.15)	0 (0.77)		11 (2.11)			
	−4 (0.79)	0 (0.22)			7 (1.76)		
	−6 (1.02)	0 (0.64)				11 (2.00)	
	−7 (1.25)	0 (0.55)					12 (2.35)
Consumption risk sharing	24 (6.21)	−1 (2.99)	−3 (1.36)				
	27 (8.79)	−1 (3.37)		0 (0.03)			
	27 (9.04)	−1 (2.72)			6 (1.67)		
	27 (8.88)	−1 (3.41)				0 (0.01)	
	27 (8.93)	−1 (3.45)					4 (0.81)

Note: Values in parentheses are absolute t-values. Euroarea sample for the period 1987–2017. For income and consumption risk sharing, the reported coefficients are $100 \cdot (1 - \kappa_0)$, $100 \cdot (-\kappa_1)$, and $100 \cdot (-\kappa_2)$. Country-fixed effects included.